



VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2013
Higher 2

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CANDIDATE NAME

CLASS

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INDEX NUMBER

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BIOLOGY

9648/02

Paper 2 Core Paper

16 September 2013

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your CT GP/ INDEX no. and name on all the work you hand in.

Write in dark blue or blue pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions

Section B

Answer any **one** question.

Write your answers on separate answer paper provided.

At the end of the examinations,

1. hand in section A and the 2 questions you attempted from section B separately;
2. fasten all your work securely;
3. enter the number of the section B questions you have answered in the grid opposite.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

Section A: Structured questions

Answer **all** questions. All answers must be written on the spaces provided and nowhere else.

- 1 (a) Distinguish between sister chromatids and homologous chromosomes.

[2]

Fig. 1.1 shows a structure (X) being observed during a particular stage of nuclear division in a plant cell.



Fig. 1.1

- (b)(i) Comment on the significance of the formation of structure X.

[2]

Fig. 1.2 shows the changes in the amount of DNA in a plant cell as it undergoes the nuclear division.

- (ii) Use an arrow to indicate in **Fig.1.2** below where structure X may be seen. Label the arrow with the letter C. [1]

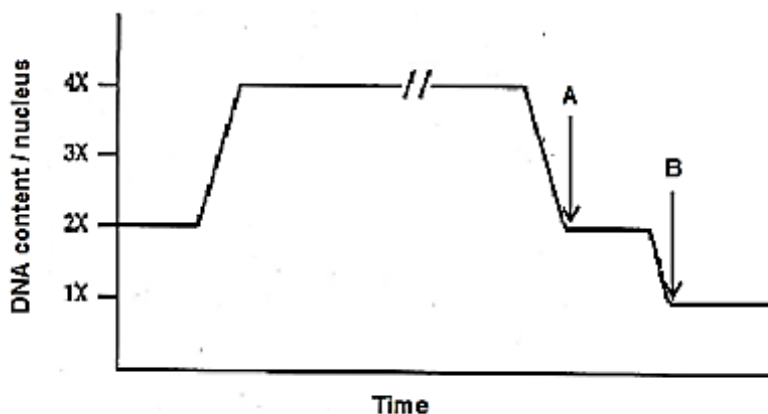


Fig. 1.2

(c) Explain why this type of nuclear division is important for the continued existence of this plant species.

[3]

(d) State one similarity and one difference between the chromosomes in stage A and B.

Similarity:

Difference:

[2]

[Total: 10 marks]

2 **Fig.2.1** below shows DNA replication in an organism

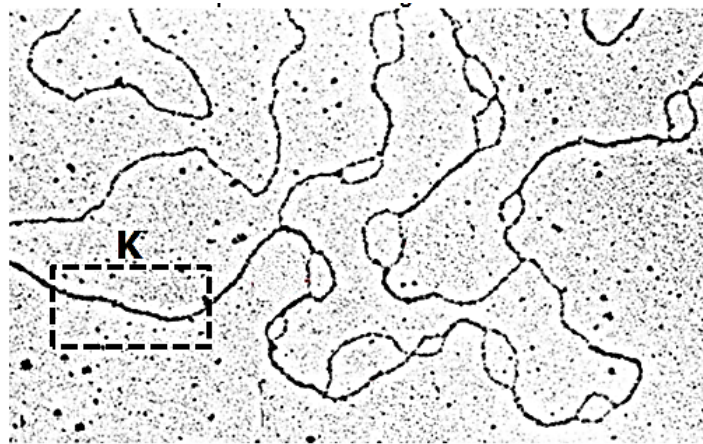


Fig. 2.1

(a) With reference to **Fig. 2.1**, state, with a reason, if this organism is eukaryotic or prokaryotic.

[1]

(b) (i) The DNA replication at each replication fork is sometimes described as '*asymmetrical*' replication as there are differences in the way the daughter strands are being synthesized. State two such differences.

[2]

(ii) Suggest a reason for these differences in (b) (i).

[1]

(c) Explain the role of proteins in the region marked 'K'.

[2]

Messelson and Stahl performed an experiment in 1957 to verify the semi-conservative replication of DNA. They first cultured the bacterium *Escherichia coli* in a medium containing ^{15}N for several generations. They then transferred the bacterial cells to a ^{14}N medium and the cells were allowed to divide once. DNA was then extracted from these cells and subjected to density gradient centrifugation. Their results are shown in **Fig. 2.2**.

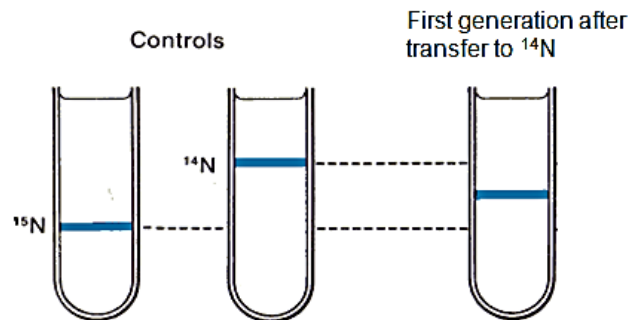


Fig. 2.2

(d) Describe and explain the next step of the experiment that they carried out to prove that DNA replicates semi-conservatively.

[4]

[Total:10 marks]

- 3 In bacteria, genes may be organized into operons. An operon with three genes **A**, **B** and **C** has been discovered recently and its control in the presence of a small molecule **S** is as shown in **Fig. 3.1**.

Small molecule **S** present:

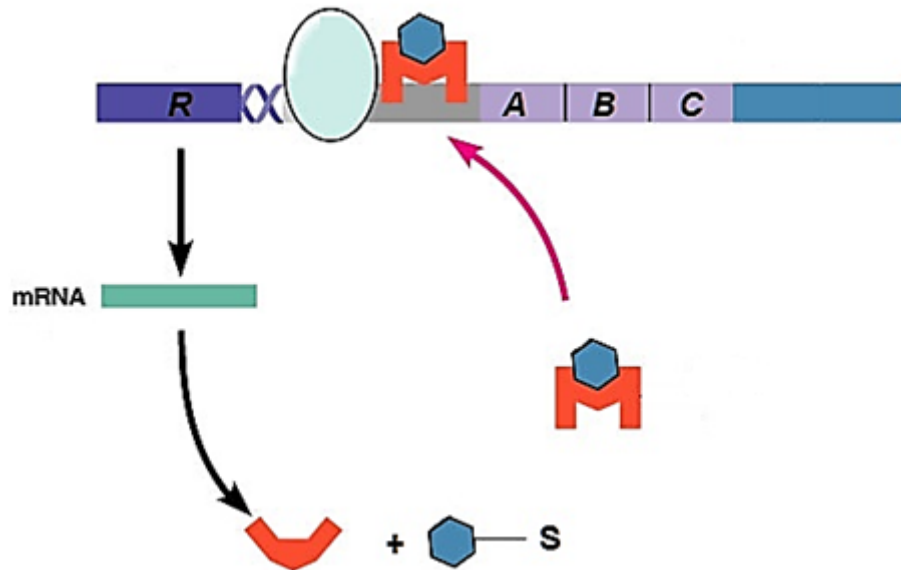


Fig. 3.1

- (a) With reference to **Fig. 3.1**,
(i) explain what is an operon.

.....
.....
..... [2]

- (ii) explain how a single mRNA results in the production of enzymes **A**, **B** and **C**.

.....
.....
..... [2]

(iii) explain the significance of molecule S in the regulation of the operon.

[3]

Fig. 3.2 shows the molecular interactions between the repressor and the operator in the presence of the small molecule **S**.

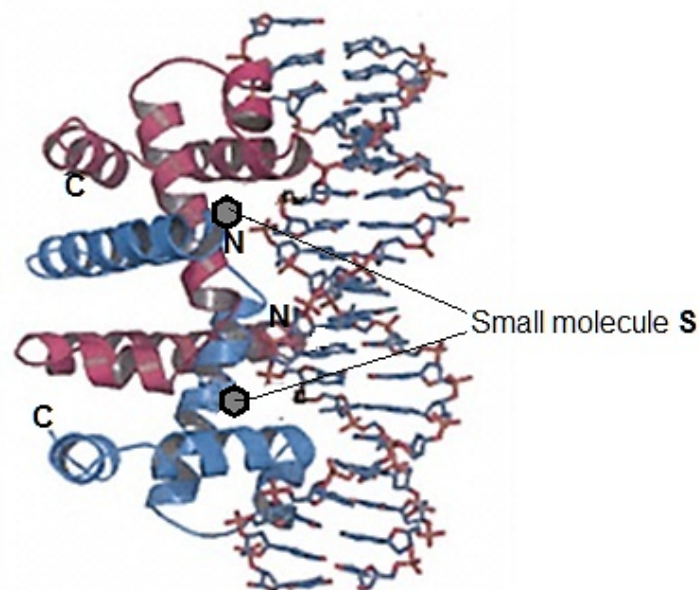


Fig. 3.2

(b) With reference to **Fig. 3.2**,
(i) describe the structure of the repressor.

[3]

(ii) explain why the active repressor only binds to the operator of the operon.

[3]

[Total:13 marks]

4 (a) (i) Describe briefly the process of transcription.

[3]

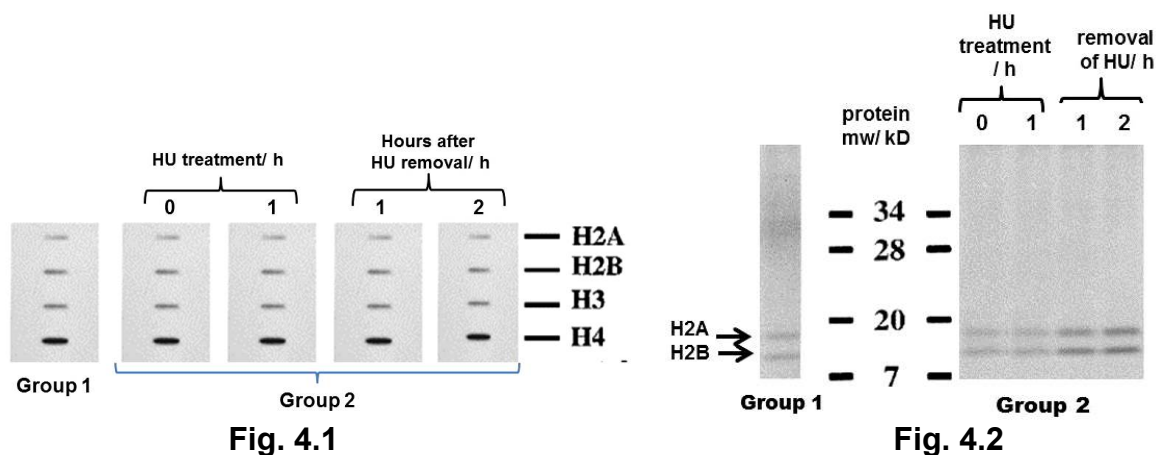
(ii) Describe two ways in which control of gene expression in eukaryotes and prokaryotes are different at the transcriptional level.

[4]

Leishmania is a unicellular parasite that causes leishmaniasis leading to death in humans. As it reproduces asexually in large numbers, the parasite cells can be used for studying the control of histone gene expression. Hydroxyurea (HU) is a drug that inhibits DNA synthesis, thereby preventing cell cycle progression.

Two groups of *Leishmania* parasites are used. **Group 1** is the untreated control. **Group 2** is incubated in a culture medium with HU for 1 hour before being transferred to a fresh medium without HU so that histone synthesis can resume. Transcription is measured by extracting radioactively-labelled histone mRNAs from parasite cell nuclei at different time points and hybridizing to single stranded histone DNAs that have been spotted onto a nitrocellulose membrane. The intensity of the radioactive signal (dark spots) is an indicator of the degree of hybridization between mRNA and the single stranded DNA (**Fig. 4.1**).

In another experiment, the same treatments are carried out to determine histone synthesis by incubating parasite cells in a mixture of amino acids containing methionine that has been labelled with radioactive isotope sulfur. Histone synthesis is measured by the intensity of the dark bands shown in the autoradiograph. **Fig 4.2** is an autoradiograph showing the levels of H2 histone proteins produced.



- (b) With reference to **Fig. 4.1** and **Fig. 4.2**,
 (i) describe the effect of HU on histone synthesis.

[2]

(ii) state the level at which histone synthesis is regulated after HU is removed. Account for your answer.

[3]

(c) Suggest how an understanding of the regulation of histone gene expression in part (b)(ii) can be used to develop strategies to control leishmaniasis.

[2]

[Total:14 marks]

- 5 A rancher owns a bull with many desirable characteristics including absence of horn. Unfortunately, it also has a sex-linked trait that in the recessive form leads to no pigment formation in the iris of the eye. This makes the bull very sensitive to sunlight and could lead to blindness.

(a) Explain how the genotype of the bull is linked to its sensitivity to sunlight.

[2]

The rancher wishes to breed the bull to a cow that will minimize the chances of any offspring showing this trait. He wants to produce another bull with most of his sire's ("father") desirable qualities but without the non-pigmented eye. A cow with absence of horn and the dominant normal coloured eyes is available for mating. However, the mating on several occasions results in 20 offspring. All of the offspring have normal coloured eyes and 4 are with horns.

(b) Using suitable symbols draw a genetic diagram to explain the above cross. [4]

Identical (monozygotic) twins arise from a single fertilized egg which divides subsequently to produce two embryos. Fraternal (dizygotic) twins, on the other hand, arise from fertilization of two eggs separately by two different sperms.

Table 5.1 shows the mean differences for certain physical traits in identical and fraternal twins reared together, and in identical twins reared apart.

Physical trait	Mean differences		
	Identical twins reared together	Fraternal twins reared together	Identical twins reared apart
Height	1.7 cm	4.4 cm	1.8 cm
Weight	1.86 kg	4.54 kg	4.50 kg
Head length	2.9 mm	6.2 mm	2.2 mm
Head width	2.8 mm	4.2 mm	2.8 mm

Table 5.1

(c) (i) What name is given to the type of variation of which height and weight are examples?

[1]

(ii) Using data from Table 5.1, evaluate the importance of genes and environment on the expression of height and weight.

[4]

[Total:11 marks]

- 6 **Fig. 6.1** below shows how insulin secretion in a β -cell is triggered by changes to blood glucose levels.

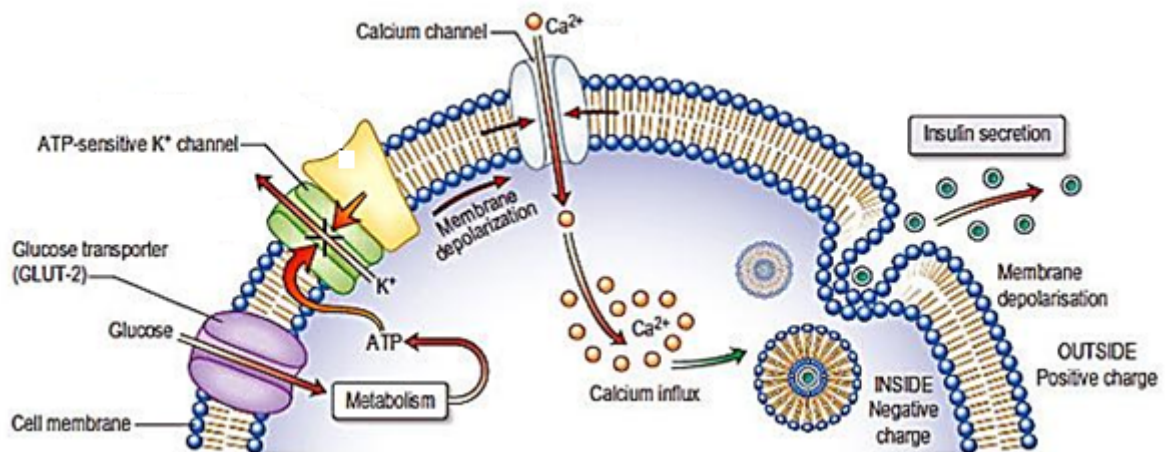


Fig. 6.1

(a) With reference to **Fig. 6.1**, describe how the increase in ATP:ADP ratio in the cell can be used as a trigger for insulin secretion.

[3]

(b) Outline the role of organelles involved in the events that lead to the embedding of glut-2 transporter in the plasma membrane.

[4]

(c) Explain why insulin is regarded as a hormone.

[2]

When insulin reaches the liver cell, it will trigger off a series of events within the cell.

(d) With reference to cell signaling, describe the main differences between the events occurring at the liver cells with those observed in **Fig. 6.1**.

[3]

[Total:12 marks]

- 7 Myxoma virus causes a disease of rabbits called myxomatosis. In the early 1950s, this virus was used to control rabbit populations in Australia. The virus is widespread amongst Brazilian rabbits (found in South America) which suffer only a slight skin disorder. In European rabbits (found in Europe and Africa and introduced into Australia), the virus rapidly leads to a fatal disease. Following infection, death occurs within approximately two weeks. In Australia, the myxoma virus can be spread from rabbit to rabbit by mosquitoes.

A study of the infection was carried out to determine the reason for the decreasing effectiveness of the disease as a means of biological control. A single strain of the virus was used throughout the whole study. Each year, numbers of uninfected rabbits were caught and injected with the virus strain.

Fig.7.1 shows the changes in the percentage of resistant rabbits in Australia that are susceptible to the myxoma virus from 1950 to 1956.

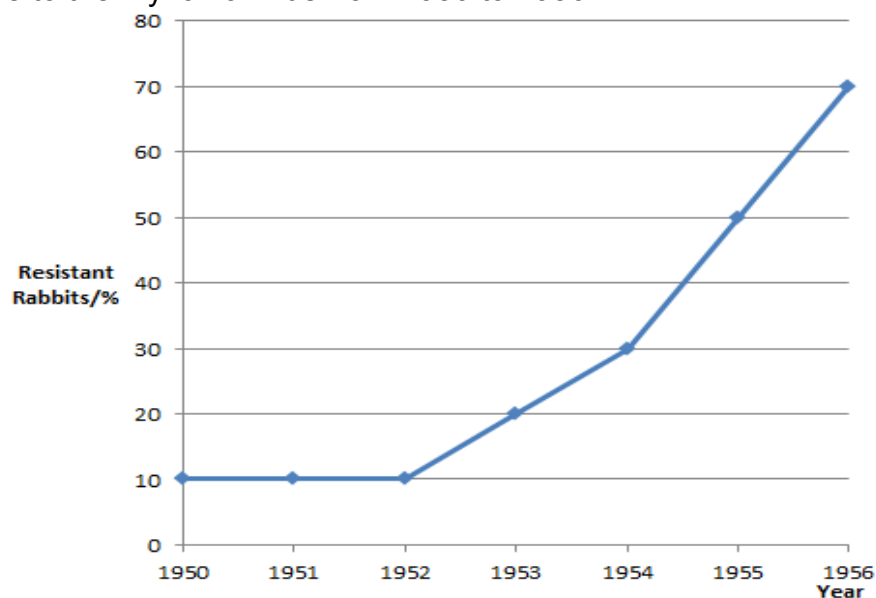


Fig. 7.1

(a) Suggest how this resistance in the rabbits may arise and spread in the population.

[4]

At the same time the virulence of the myxoma virus was also determined. Virulence is defined as the ability of the virus to infect and kill their hosts. **Fig 7.2** shows the virulence of the myxoma virus over the same period of time.

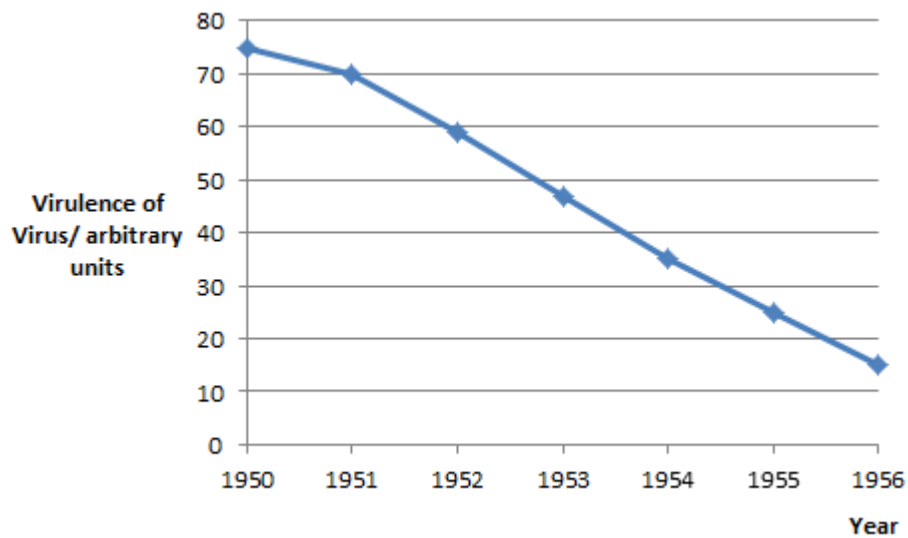


Fig 7.2

(b) Using the data in **Fig 7.1** and **Fig. 7.2**, discuss the significance of the change in virulence to the myxoma virus.

[3]

To study the evolution of the myxoma virus in Australia, the DNA sequence of the genes involved in virulence was sequenced from the various samples of myxoma virus taken from the rabbit population. **Fig 7.3** below shows the phylogenetic tree that was constructed based on these samples. A strain of the virus from Europe, where the virus was also found, was included as a comparison.

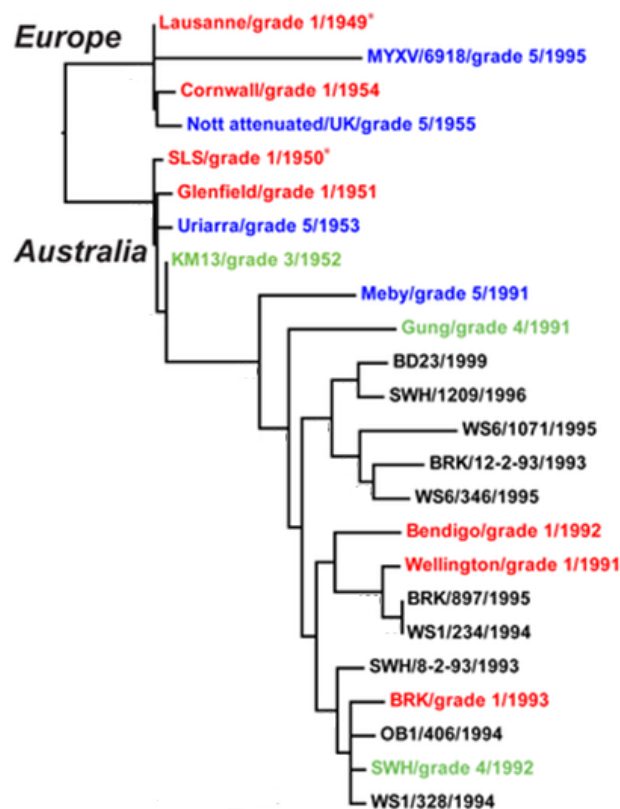


Fig 7.3

(c) (i) What do you understand by the term phylogenetic tree?

[2]

(ii) Suggest a reason for the large number of different strains of myxoma virus in Australia compared to Europe.

[1]

[Total:10 marks]

Section B

Answer **one** question only.

Write your answers on separate writing paper.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answer must in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

Begin your answers to each part (a), (b) and (c) on a new sheet of writing papers.

8. (a) Compare and contrast triglyceride and starch as storage compound. [8]
- (b) Explain how anatomical and embryological homology can be used to support Darwin's theory of Natural Selection. [6]
- (c) Discuss the significance of telomerase in different types of cells. [6]
9. (a) With named examples, describe the process and significance of gene amplification in cells. [6]
- (b) Explain how genetic variation may be preserved in a population. [8]
- (c) Discuss the importance of membranes in the reproductive cycle of the influenza virus. [6]